Prototyping Area

Draw your project schematic here

Headers / Solder Pads

Power
H2
RESET
3V3D
5V
GND
GND
VIN

Analog
H3
A0
A1
A2
A3
A4_SDA
A5_SCL

C1
100nF 0603
GND

D10, 11, 12, 13 used by Ethernet interface.
D2 used if WIznet Ethernet I/C 'Interrupt' jumper enabled.
D4 used by microSD card Chip Select.

H1
A5_SCL
A4_SDA
AREF
GND
D13_SCK
D12_MISO
D11_MOSI
D10_SS
D9
D8
M100.1

D7
D6
D5
D4 uSD_nCS
D3
D2
D1_TX
D0_RX
H4

Headers / Solder Pads

3V3 Power Supply 800mA

REG1 LD1117-SOT89
Dual Footprint
REG2 LD1117-SOT223
5V
3V3D
C9 1uF 0603
C10 10uF 6.3V A TANT
GND

1.8V / 3.3V Decoupling

L1 1uH 250mA 0805
1V8_OUT
1V8A
C3 10uF 6.3V A TANT
C4 10uF 6.3V A TANT
L2 1uH 250mA 0805
3V3D
3V3A
C13 10uF 6.3V A TANT
GND

Wiznet W5100 Power-On/Reset

APX81X-RESET-IC
WIZRST
RESET
APX811-44UG-7 Reset IC
140ms min, 280ms max
< 4.38 Vrst, push-pull output

X2 25MHz
R3 1M
C15 13pF
C16 13pF

Status Display

WTXLED RN1A 1K LED1 YELLOW
WRXLED RN1B 1K LED3 YELLOW
COLLED RN1C 1K LED4 RED
FDXLED RN1D 1K LED7 GREEN
SPDLED RN2C 1K LED8 GREEN
LINKLED RN2D 1K LED9 YELLOW

Ethernet Module with PoE

HY931147C-RJ45
Green LED
Yellow LED
RN2A 1K FDXLED
RN2B 1K LINKLED
C18 10nF
J1/T+ J2/T- J3/R+ J4 J5 J6/R- J7 J8
1CT:1
1CT:1
RN5A 49R9 1%
RN5B 49R9 1%
RN5C 49R9 1%
RN5D 49R9 1%
C22 100nF
C23 100nF
C24 100nF
RXIP RXIN TXOP TXON
WIZRST INT
SJ4 D2_INT D2

MNT1 MNT2 FT-POE-MODULE3
PoE supply feed to Arduino VIN –>
VIN
For DC PoE feed, jumper "+ to VIN", "– to GND" directly

RN7D 1K
Q3 MOSFET-NCHANNELSMD
D10_SS
D12_MISO
D11_MOSI
D10_SS
D13_SCK
SEN
SPI MISO tri-state by SS/nCS
SPI D10_SS/nCS low = SEN high = SPI enabled

Freetronics EtherTen
www.freetronics.com/et

DC Regulated Power Supply

D1 1N4004 SMB
VIN
C5 47uF 25V
IC1 MC33269D-5
C7 100nF 0603
C6 47uF 25V
5V
LED2 PWR/BLUE
RN6A 1K

DCInput/USBVCC Auto-switch

VIN
RN4C 10K
RN4D 10K
3V3D
USBVCC
Q1 MOSFET-PCHANNEL
C14 100nF 0603

In Circuit Serial Programming

P2 ICSP:MAIN
D12_MISO
D13_SCK
RESET
D11_MOSI
GND

P3 ICSP:USB
MISO2
SCK2
RESET2
MOSI2
GND

micro SD Card Slot

USD-SUCKETNEW
D4 uSD_nCS
D11_MOSI
D13_SCK
D12_MISO
RN6D 1K
RN6C 1K
RN6B 1K
RN8B 2K2
RN8C 2K2
RN8A 2K2
C29 100nF 0603
CD0 M01PTH

ATmega16U2 Microcontroller

RN4A 10K
RESET2
R1 1M 0603
C2 22pF 0603
X1 16MHz
C8 22pF 0603
R2 27R 0603
C11 1uF 0603
C12 100nF 0603
RESET
XTAL2
XTAL1
AVCC
VCC
GND
UCAP
UVCC
D-
D+
UGND
PAD
ATMEL_ATMEGA8U2-MUQFN

MISO2 MOSI2 SCK2

TX/YEL LED5 RX/GRN LED6
RN3A 1K RN3B 1K

USB-6PSMT
F1 500mA PTC 1812
PGB1010603 PGB1010603
0805 INDUCTOR
R4 22R 0603
R5 22R 0603
SJ1 GROUND (Linked)
Short across SJ1 GROUND and SJ3 RESET-EN after all other routing

(CTS)PD7
(RTS)PD6
(XCK)PD5
(TXD1)PD3
(RXD1)PD2
TXLED
RXLED
USBRXD
USBTXD

ATmega328P Microcontroller

IC5
UBOOT SJ2
RESET-EN SJ3
C19 100nF 0603
RN4B 10K
S1 RESET
C20 22pF 0603
C21 22pF 0603
X3 16MHz
R7 1M 0603
AVR-MEGA8-AI

RN3C 1K
RN3D 1K
M01PTH A6
M01PTH A7
D10 LED
R8 1K 0603
Q2 MOSFET-NCHANNELSMD

PC0(ADC0) A0
PC1(ADC1) A1
PC2(ADC2) A2
PC3(ADC3) A3
PC4(ADC4/SDA) A4_SDA
PC5(ADC5/SCL) A5_SCL
D0_RX
D1_TX
D2
D3
D4 uSD_nCS
D5
D6
D7
D8
D9
D10_SS
D11_MOSI
D12_MISO
D13_SCK

C27 100nF 0603
I2C Addr: 70 (0x50)
R14 4K7
R15 4K7
A4_SDA
A5_SCL
SJCN
SJCN
MCP24AA025E48
RN8D 2K2
RN7C 1K
RN7A 1K

Sheet: /
File: working.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. eeschema 7.0.6
Id: 1/1